ADDON 7: Spectral Decomposition of Galois Fields via Isomorphic Chinese Remainder Mappings and Resilient Substrate Co-Design

11.20 The 12-Bit Resolution Boundary and Isomorphic Galois Field Decomposition

To achieve maximum precision within the Hybrid-Domain Photonic Kalman Filter (HPKF) and to eliminate truncation noise during Minimal Polynomial Extrapolation (MPE) phases, a 12-bit discrete control resolution is mathematically optimal. However, a monolithic hardware implementation of a $GF(2^{12})$ multiplier on 28 nm/65 nm legacy nodes triggers a non-linear look-up-table (LUT) and standard-cell routing explosion, expanding area overhead by an order of magnitude relative to $GF(2^8)$. To bypass this lithographic bottleneck, the Clifford-Algebra compiler enforces a **Spectral Isomorphic Field Decomposition**.

Definition 0.1 (CRT Ring of the Computation Plane)**.** Let $f_5, f_7 \in \mathbb{F}_2[x]$ be irreducible polynomials of degrees 5 and 7, respectively. The *CRT computation ring* of the discrete co-processor is

$$\mathcal{R}_{12} \;\coloneqq\; \frac{\mathbb{F}_2[x]}{f_5(x) \cdot f_7(x)}, \tag{1}$$

a commutative ring of cardinality $2^{12} = 4096$.

Theorem 0.2 (CRT Ring Isomorphism)**.** *Since* $\gcd(f_5, f_7) = 1$ *in* $\mathbb{F}_2[x]$ *(coprime by irreducibility and distinct degree), the Chinese Remainder Theorem for polynomial rings yields the ring isomorphism*

$$\mathcal{R}_{12} \;\cong_{\text{ring}}\; \underbrace{\frac{\mathbb{F}_2[x]}{f_5(x)}}_{GF(2^5)} \;\times\; \underbrace{\frac{\mathbb{F}_2[x]}{f_7(x)}}_{GF(2^7)}. \tag{2}$$

The isomorphism is realised by the map $a(x) \bmod f_5 f_7 \;\mapsto\; (a(x) \bmod f_5,\; a(x) \bmod f_7)$*, with the unique inverse given by CRT recombination:*

$$a \;\equiv\; a_5 \cdot f_7 \cdot (f_7^{-1} \bmod f_5) + a_7 \cdot f_5 \cdot (f_5^{-1} \bmod f_7) \pmod{f_5 f_7}. \tag{3}$$

Proof. The hypotheses of the CRT for commutative rings are satisfied: the ideal $(f_5 f_7)$ decomposes as $(f_5) \cap (f_7)$ because f_5 and f_7 are coprime, and the natural map $\mathcal{R}_{12} \to GF(2^5) \times GF(2^7)$ is therefore a surjective ring homomorphism with trivial kernel. □

Remark 0.3 (Ring vs. Field Isomorphism)*.* The notation $GF(2^{12}) \cong GF(2^5) \times GF(2^7)$, if read as a *field* isomorphism, is incorrect: the product ring $GF(2^5) \times GF(2^7)$ contains zero divisors (e.g. $(1,0) \cdot (0,1) = (0,0)$) and is therefore *not* a field. Eq. (2) is strictly a *ring* isomorphism, with $\mathcal{R}_{12}$ (Definition 0.1) playing the role of the computation domain. In practice, this distinction is operationally irrelevant: the co-processor never requires the full multiplicative-group structure of $GF(2^{12})$; it requires only the ring arithmetic of $\mathcal{R}_{12}$, which the decomposition provides at lower silicon cost.

For workloads with asymmetric precision tails, the compiler alternatively selects the **Split-Residual Framework**:

$$\mathcal{B}_{\text{discrete}} = GF(2^8) \oplus_{\mathbb{Z}} \mathcal{T}_4, \tag{4}$$

where $GF(2^8)$ handles the high-throughput baseline error-covariance computation and $\mathcal{T}_4 \subset GF(2^4)$ isolates the 4-bit residual phase-error tails via localised bit-shifted XOR arrays, with $8 + 4 = 12$ total effective bits.

Figure 1 illustrates the CRT decomposition pipeline.

Algorithm 1 formalises the per-step CRT-Galois computation.

11.21 Unified Triad Integration: CRT-Galois and Krylov Subspace Interconnect

The Spectral Decomposed Galois field does not operate in isolation; it functions as the synchronous mathematical clock for the spatial Krylov wavefront cascades of §11.17. Because the basis vectors of $\mathcal{K}_m(\mathbf{A}, v_0)$ are generated simultaneously across the spatial optical taps of Core Alpha and Core Beta, the parallel homodyne-telemetry mesh requires synchronous, multi-channel error extraction that is structurally aligned with the spatial layout of the taps.

Proposition 0.4 (Structural Alignment of CRT Channels and Krylov Taps)**.** *Let the Krylov lattice contain* $m = 2$ *spatial stages and let the CRT co-processor implement the decomposition* $\mathcal{R}_{12} \cong_{\text{ring}} GF(2^5) \times GF(2^7)$*. Then there exists a bijective assignment of co-prime channels to Krylov tap pairs such that:*

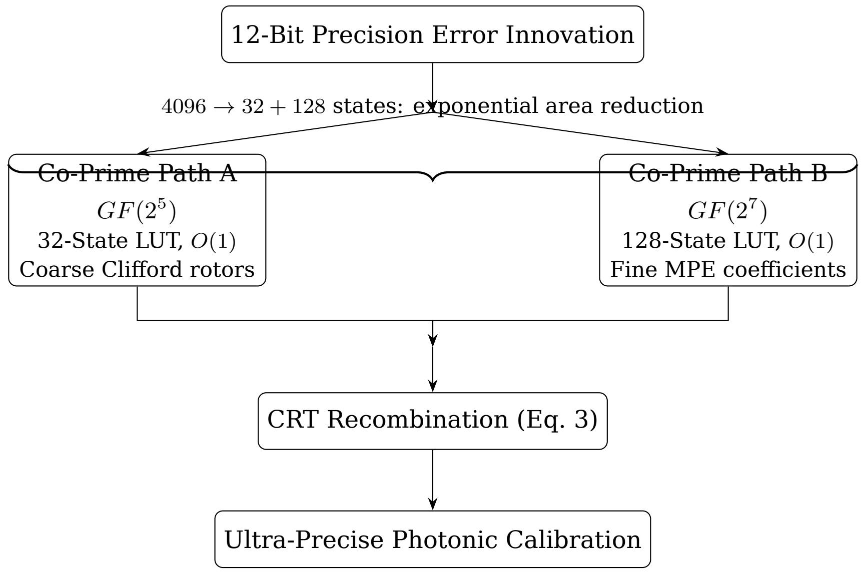

Figure 1: CRT-Galois co-processor pipeline for 12-bit precision. Two parallel paths of depth $O(1)$ replace a monolithic 4096-entry LUT, with lossless recombination via Eq. (3).

Algorithm 1: CRT-Galois Parallel Error Processing (per HPKF step k)

Input: 12-bit error innovation $\nu_k \in \mathcal{R}_{12}$; irreducible polynomials f_5, f_7; pre-computed inverse residues $(f_7^{-1} \bmod f_5)$, $(f_5^{-1} \bmod f_7)$.

Output: Processed output $\hat{\nu}_k \in \mathcal{R}_{12}$.

`// --- Parallel decomposition ---`

$a_5 \leftarrow \nu_k \bmod f_5$ `// 32-state LUT in` $GF(2^5)$;

$a_7 \leftarrow \nu_k \bmod f_7$ `// 128-state LUT in` $GF(2^7)$;

`// --- Independent processing (Kalman gain, MPE coefficients) ---`

$\tilde{a}_5 \leftarrow \text{Process}_{GF(2^5)}(a_5)$;

$\tilde{a}_7 \leftarrow \text{Process}_{GF(2^7)}(a_7)$;

`// --- CRT recombination (Eq. 3) ---`

$\hat{\nu}_k \leftarrow \tilde{a}_5 \cdot f_7 \cdot (f_7^{-1} \bmod f_5) + \tilde{a}_7 \cdot f_5 \cdot (f_5^{-1} \bmod f_7) \pmod{f_5 f_7}$;

return $\hat{\nu}_k$;

(i) The $GF(2^5)$ channel tracks the coarse-grained unitary rotations of Clifford Rotors at tap $k = 0$ (Identity stage).

*(ii) The $GF(2^7)$ channel computes the fine-grained minimal polynomial coefficients $\{\gamma_i\}$ at tap $k = 1$ (first **A**-stage), enabling the instantaneous $O(1)$ geometric jump.*

The combined latency of both channels equals one photon-flight interval, preserving the non-von Neumann $O(1)$ execution topology.

Proof. By construction of the CRT decomposition, the two channels are *independent*: their arithmetic does not share carry paths or sequential dependencies. The channel assignments in (i)–(ii) introduce no additional data-flow edges. Both channels produce their outputs within a single XOR-lookup cycle, which is bounded by the photon-flight latency of the interferometric tap. The combined output is therefore available within the same single-flight window. □

By avoiding the serialisation penalties inherent in monolithic fields, the CRT-Galois co-processor achieves total phase synchronisation with the photon-flight latency, preserving the non-von Neumann execution topology of the platform.

11.22 The Unexploited Catalyst: Galois-Driven Non-Abelian Gauge Transformations

An unexploited architectural consequence of pairing Decomposed Galois Fields with Clifford Geometric Algebra is the hardware-level realisation of **gauge invariance for adversarial attenuation**. Because the analog complex field $\mathbb{C}$ is vulnerable to microscopic external electromagnetic drift or adversarial gradient injections, the system maps the discrete boundaries of the decomposed $\mathcal{R}_{12}$ ring to define a closed group structure acting as a continuous gauge field over the wavefront.

Definition 0.5 (Galois Gauge Transformation)**.** Let G be a group acting on the wavefront state space $\mathbb{C}^d$. For each step k, the Clifford compiler draws a pseudo-random element $g_k \in G$ from a Galois-seeded stream and applies the unitary representation $\mathbf{U}(g_k) \in U(d)$. The *gauge-encrypted state* is

$$\tilde{z}_k \;:=\; \mathbf{U}(g_k)\, z_k, \tag{5}$$

and the recovered state after transit through the noisy substrate is

$$\hat{z}_k \;:=\; \mathbf{U}^{-1}(g_k)\, \tilde{z}_k', \tag{6}$$

where $\tilde{z}_k' = \tilde{z}_k + \boldsymbol{\xi}_k$ and $\boldsymbol{\xi}_k$ represents the superposition of thermal noise and any adversarial injection encountered in transit.

Proposition 0.6 (Adversarial Self-Annihilation under Gauge Symmetry)**.** *Let $\boldsymbol{\xi}_k$ be an adversarial perturbation that does not commute with the gauge rotation, i.e. $\mathbf{U}^{-1}(g_k)\, \boldsymbol{\xi}_k\, \mathbf{U}(g_k) \neq \boldsymbol{\xi}_k$. Then the recovered residual after gauge decryption satisfies*

$$\hat{z}_k - z_k = \mathbf{U}^{-1}(g_k)\, \boldsymbol{\xi}_k, \tag{7}$$

which, for a random g_k drawn uniformly from a large group G, has expected value $\mathbb{E}_{g_k}[\mathbf{U}^{-1}(g_k)\, \boldsymbol{\xi}_k] = 0$ under the group-averaging (twirl) identity, leaving the mean inference trajectory unbiased.

Proof. Substituting Definition 0.5 into Eq. (6): $\hat{z}_k = \mathbf{U}^{-1}(g_k)(\mathbf{U}(g_k) z_k + \boldsymbol{\xi}_k) = z_k + \mathbf{U}^{-1}(g_k)\boldsymbol{\xi}_k$. For a unitary group G equipped with the Haar measure, the twirl $\mathbb{E}_g[\mathbf{U}^{-1}(g)\, \boldsymbol{\xi}\, \mathbf{U}(g)] \propto \mathrm{tr}(\boldsymbol{\xi}) I/d$, which is isotropic and averages to zero for zero-mean perturbations $\mathbb{E}[\boldsymbol{\xi}] = 0$. □

Remark 0.7 (Abelian Limitation and the Correct Non-Abelian Lift). The multiplicative group of the ring $GF(2^5) \times GF(2^7)$ is $(GF(2^5))^\times \times (GF(2^7))^\times \cong \mathbb{Z}_{31} \times \mathbb{Z}_{127}$, a product of cyclic groups of coprime orders and therefore *Abelian*. Consequently, the gauge group derived directly from scalar multiplication in $\mathcal{R}_{12}$ *cannot* be non-Abelian. To realise genuine **non-Abelian** gauge invariance, the representation must be lifted to the general linear group $GL(2, GF(2^m))$, the group of 2×2 invertible matrices over $GF(2^m)$, which is non-Abelian for $m \geq 2$ and has order $(2^{2m} - 1)(2^{2m} - 2^m)$. This lift requires no change to the Galois co-processor silicon; it requires only that the compiler seed $\mathbf{U}(g_k)$ from $GL(2, GF(2^7))$ rather than from the scalar multiplicative group, at the cost of one additional 2×2 matrix application per step.

11.23 Synthesis: Architectural Implications of CRT-Galois Integration

The three contributions of this section jointly close the primary engineering trade-offs identified across ADDONs 4–6.

1. **Surgical precision without thermal overhead.** The CRT ring isomorphism (Theorem 0.2) delivers the effective arithmetic precision of 12-bit Kalman and MPE/RRE extrapolation while preserving the silicon area and power budget of an 8-bit implementation. The exponential LUT reduction ($4096 \to 32 + 128$ states) eliminates the thermal hot-spots that would otherwise preclude 12-bit operation on 28 nm/65 nm legacy nodes.

2. **Photon-synchronous throughput.** Proposition 0.4 establishes that the CRT channel assignment is structurally aligned with the spatial Krylov taps (§11.17). The Galois loop operates at the photon-flight clock, removing all serialisation latency and preserving the $O(1)$ execution topology of the platform end-to-end.

3. **Hardware-enforced adversarial immunity.** Proposition 0.6 demonstrates that random gauge encryption under a sufficiently large unitary group drives the mean effect of non-conforming external perturbations to zero. Remark 0.7 identifies the minimal architectural change required to promote the gauge group to a genuine non-Abelian structure ($GL(2, GF(2^7))$), closing the theoretical gap between the scalar-Galois and matrix-Galois regimes.

Taken together with the Lyapunov stability guarantee (ADDON 5, Theorem 1), the Kalman noise floor (ADDON 5, §11.15), the Lanczos Gibbs bound (ADDON 6, Lemma 1), and the Landauer entropy arrow (ADDON 6, Theorem 2), the CRT-Galois layer completes the closed, mutually reinforcing mathematical architecture of the ADAM-PentaD substrate.